

Odoo 17

Implementation guide

Configuration • Custom addons • Operational workflows • Acceptance testing

A complete application blueprint for self-hosted Odoo 17 Community, based on the EL MOKRIF project report.

34 pages | 12 functional areas | Detailed execution steps

Includes architecture, permissions, data setup, custom addon specifications, end-to-end pilots, migration controls and a phased delivery roadmap.

Scope: Odoo applications and business workflows. Server provisioning, Docker, DNS, HTTPS, backups and deployment are excluded.

Implementation blueprint; no live database was inspected or changed. Proposed addons are specifications, not delivered code. Tax and HR parameters require responsible company validation.

01 | How to use this guide

Reading and execution order

Read the architecture and foundation pages first. Configure one test company and one representative chantier, then execute the purchase-to-consumption pilot before broad data import. Complete sales and finance next; build the dashboard after its sources reconcile.

Each functional section distinguishes the standard foundation from extensions, gives configuration actions, describes the operational sequence and ends with an acceptance gate. Passing a gate means demonstrating the stated result with representative records and the intended user roles.

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Labels used throughout

STANDARD: configure an available Odoo 17 Community capability. **EXTENSION:** select and validate a compatible third-party addon, or implement the missing behavior. **CUSTOM:** proposed EL MOKRIF business logic. **OPTIONAL:** a useful enhancement beyond the report's explicit scope.

Do not install an addon merely because its name sounds relevant. Verify its 17.0 branch, license, dependencies, maintenance, security behavior and fit with the selected accounting/stock architecture. A listed candidate category is not a verified package recommendation.

Working records to maintain

Keep a configuration register, addon register, data-import mapping, decision log and test evidence register. For each requirement, record its owner, solution, dependencies, test and approval. A feature remains open until its behavior has been demonstrated; installation alone is not acceptance.

02 | Requirements and Community gaps

The report's 12 areas

Area	Community foundation	Additional delivery needed
Configuration	Companies, users, groups	Chantier restrictions, business approval rules
Discussion	Discuss, chatter, activities	Routing/templates and targeted automation
Calendar	Events, attendees, reminders	Provider synchronization and business links
Accounting	Invoices, entries, payments, analytics	Full reports, matching tools, follow-ups, cheque layouts
Human resources	Employees, contracts, leave, recruitment	Appraisals and local policy configuration
Documents	Business attachments	Central GED, versioning and document workflow
Contacts	Customers, vendors and addresses	Missing identifiers and data quality rules
Purchase	RFQs, POs, receipts, agreements	Site requests, strict two-person approval if required
CRM	Leads, opportunities, activities	Reliable J+3 quotation follow-up and targets
Inventory	Locations, routes, valuation, tracking	Enforced quality release, scanning interface, site consumption
Sales	Quotations, orders, portal, pricing	Chantier links; certified progress billing if selected
Dashboards	Lists, pivots and graphs	Central, permission-aware management dashboard

Corrections to the report's implementation assumptions

- A chantier is normally a project with an analytic account and stock location, not a company. Create companies for actual legal entities.
- A purchase receipt does not automatically authorize posting or paying a vendor bill. Keep accounting checks and payment approval.
- Standard replenishment rules are product/location rules. Category policies can seed defaults, but are not themselves the rules.
- A Quality location does not block material release by itself. Enforce inspection approval in business logic.
- Quotation signature is distinct from a general electronic-signature application. Barcode fields/labels are distinct from a complete warehouse scanning interface.
- Rules-based bank matching is not automatically AI. Keep the report's literal AI requirement open until a suitable solution is selected and demonstrated.

Supporting additions

Project/chantier management is necessary to structure the report's site-based analytics. Expenses, timesheets, detailed profitability, retention and situations de travaux are conditional additions. Implement them only with explicit business definitions; they must not silently replace the 12 required areas.

03 | Core data architecture

One source of truth for each business concept

Business object	Odoo model / design
Legal entity	res.company
Customer, vendor, subcontractor	res.partner
Chantier	Extend project.project
Work package	project.task
Financial allocation	account.analytic.account and analytic distribution
Site stock	Internal stock.location belonging to its company
Commercial transaction	sale.order / sale.order.line
Procurement	purchase.order / purchase.order.line
Physical movement	stock.picking / stock.move / stock.move.line
Posted financial transaction	account.move / account.move.line
Document	Attachment plus selected document-management layer

Create the chantier foundation

1. Extend project.project with a chantier flag, reference, customer, company, address, region, manager, planned dates and lifecycle state. Reuse suitable standard fields instead of duplicating them.
2. Define work-type classification: construction, carpentry and finishing. Allow task-level distinctions when a site contains several types of work.
3. Link one primary analytic account and one internal site stock location. Reuse native project analytic links where the installed modules supply them.
4. Implement an Initialize chantier action that creates only missing linked records, uses the correct company and records the action in chatter.
5. Add smart buttons for related sales, purchases, movements, invoices and documents. Counts and actions must obey access rules.
6. Allow multiple orders for the same chantier. Users must choose an existing project for additional work rather than automatically creating another project.

Lifecycle and controls

Draft → Approved → In progress → On hold → Completed → Closed. A manager approves a complete dossier. In progress permits ordinary operations. On hold blocks new commitments unless an authorized exception is recorded. Completed means physical work is finished; accounting settlement may continue. Closed requires remaining stock, open requests, orders and financial exceptions to be reviewed.

Do not delete historical chantiers. Archive them. Reopening requires a reason and an authorized role. Cancellation/closure must never delete posted accounting entries or completed stock moves.

Acceptance gate

Initialize the same chantier twice: there must still be one linked analytic account and one site location. Try linking a project, stock location or order from another company: the relationship must be rejected or prevented. Verify historical records remain accessible to authorized users after closure.

04 | Company configuration and access

Configure the foundation

1. In Settings > Users & Companies > Companies, create/verify EL MOKRIF COMPANY SARL. Confirm the report's address, ICE 000070313000033 and RC 18695 Marrakech with the company before use.
2. Set country to Morocco and currency to MAD before accounting transactions. Configure document logo, address and legal footer.
3. Install French and Arabic; set each user's language and Africa/Casablanca time zone. Set customer language for generated communications.
4. Create additional companies only for confirmed legal entities. Decide which contacts/products can be shared and which remain company-specific.
5. Create named users, enable two-factor authentication and assign the least privileges needed for their duties. Keep configuration permissions separate from business management permissions.
6. Test with one account per operational role. Do not use the administrator account for acceptance testing.

Role	Allowed business actions	Restrictions
Management	View results, approve designated actions	No routine technical administration
Accountant	Post entries, review bills, reconcile	Payment authority according to policy
Salesperson	Own/team leads, quotations and customers	No purchase or journal approval
Buyer	Requests, RFQs, supplier comparison	No self-approval where segregation applies
Purchase approver	Approve in assigned company/scope	Approval limits enforced in Python
Storekeeper	Receive, transfer, count	No quality override or invoice posting
Quality approver	Inspect, pass/reject, release	No silent modification of posted receipts
Chantier manager	Assigned projects and requests	No unrelated sites or HR private data
HR manager	Contracts, leave, reviews	Confidential workspace access
Employee	Own requests and permitted documents	No colleagues' private records

Implement controls

Use model access rights for broad operations and record rules for company, team and ownership scope. Use company_id, suitable check_company relationships and consistent company context in custom code. A hidden button is not security: enforce approval rights and state transitions on the server.

Employees need creation/editing of their own drafts, not read-only access, to submit leave and expenses. Restrict edits once approved. Test direct links, export, search, smart-button counts, attachments and portal access, including users allowed in several companies.

Acceptance gate

Demonstrate permitted actions and denied actions for each role. Save the configuration and test evidence. Do not proceed with real HR or commercial data until confidentiality tests pass.

05 | Contacts and product master data

Contacts: configuration and import

1. Install Contacts. Export a small sample template to confirm the field names available in this database.
2. Clean source lists: normalize names, remove obvious duplicates, verify ICE/contact identifiers, emails and addresses. Do not merge distinct legal entities just because their names resemble each other.
3. Create tags: Customer, Supplier, Subcontractor, Developer and Individual. A partner may hold several roles.
4. Create a company contact with child contact persons, invoice addresses and delivery addresses where necessary.
5. In Sales & Purchase settings on the contact, assign salesperson, pricelist, payment terms and other applicable properties. Accounting verifies financial settings.
6. Add missing ICE/RC fields through a small extension only if suitable localization fields are absent. Make required-field checks appropriate to the contact type.
7. Import with stable external IDs. Keep a source-ID-to-Odoo-ID mapping and a rejected-row correction log.

Contact workflow

Sales/buyer creates draft master data → designated owner checks identity and duplication → accountant checks payment/fiscal settings → the contact is reused in CRM, orders and invoices. Restrict sensitive changes such as payment details through the company's agreed review process.

Product master required by sales, purchase and stock

1. Define categories such as Cement, Steel, Timber, Finishing Materials, Equipment and Services.
2. Set stock-managed physical materials as storable products in Odoo 17. Use Service for labor/subcontracted work. Use consumable classification only where stock tracking is intentionally unnecessary.
3. Define base unit and purchase unit with valid same-category conversions. Test bags, kilograms, meters and units. Do not assume every bag weighs the same across products.
4. Add supplier prices, vendor product references, lead times and minimum quantities.
5. Set sales/purchase taxes, accounts, costing method and valuation at the appropriate product/category level.
6. Assign lot or serial tracking only where operationally justified. Set unique barcodes and internal references.
7. Define invoicing policies separately for materials and services. Confirm routes on a small product sample before mass import.

Acceptance gate

One business serving as both customer and supplier has one partner record. A sample PO using the purchase unit receives the correct stock quantity in the base unit. A service sale creates no unintended stock move. Updating imports by external ID changes records without duplicating them.

06 | Accounting configuration

Foundation and dependency order

Use account, analytic, l10n_ma and stock_account as appropriate. Community has core accounting models, but the full reporting/reconciliation experience described by the report needs an addon selection exercise. Validate the exact installed feature set before promising a menu or automation.

1. Install/configure the Moroccan localization before posting operational entries. If the database already has transactions, assess existing accounts and taxes first; do not blindly replace the localization.
2. Have the accountant approve the chart of accounts, receivable/payable accounts, revenue, material/service expenses, stock valuation and interim accounts.
3. Create Sales, Purchases, Bank and Cash journals. Create separate bank journals for the actual CIH/Attijariwafa accounts used, if confirmed.
4. Configure journal defaults, numbering, bank details, fiscal year and lock dates. Restrict posting and unlocking rights.
5. Configure payment terms, including actual due-date rules and installments. Set customer/vendor defaults.
6. Validate applicable taxes and fiscal positions against current company obligations. The PDF's 20%, 14%, 10% and 7% list is not a verified current tax schedule.
7. Configure rounding, invoice lines, credit notes and down-payment accounts with the accountant. Confirm legal identifiers on document templates.
8. Configure inventory category accounts and automated valuation consistently before importing opening stock.
9. Enable the selected analytic dimensions and mandatory allocation controls for relevant revenue/expense lines.
10. Verify a sample invoice, bill, credit note and payment before importing accounting data.

Posting workflow

Draft documents may be corrected by authorized preparers. An accountant reviews partner, dates, quantities, accounts, taxes, due dates and analytics, then posts. Posted corrections use the supported reversal/credit-note mechanism. Do not edit historical journal data directly in SQL.

Required reports

General ledger, trial balance, aged receivable/payable and journals must be available through verified Community-compatible reporting extensions if absent. Configure consistent company/date filters. Include opening balances and posted/draft treatment in every report definition.

Acceptance gate

Post one customer invoice and one supplier bill. Confirm journal entry balance, taxes, due dates, analytic allocation and partner balances. Create a partial payment and credit note; confirm residual amounts. Reconcile the trial balance and general ledger for the same date range. An unauthorized user must not post or unlock periods.

07 | Analytic dimensions and cost policy

Configure dimensions

1. Create a Chantier analytic plan and one analytic account per site. Ensure its company agrees with the project.
2. Create a Type of Work plan when finance requires cross-analysis by construction, carpentry and finishing. Define Region as a project field; add a separate analytic plan only if financial cross-reporting needs it.
3. Define applicability rules so relevant project revenue and expense lines require allocation. Exclude unrelated balance-sheet, tax and overhead lines where appropriate.
4. Configure distribution defaults from chantier/context. Use server-side checks for missing or inconsistent allocations in imports and automated creation.
5. For shared bills, split at line level or use approved analytic distribution percentages. Verify 100% allocation within the relevant plan; do not confuse independent dimensions with an expense split.
6. Test propagation from order line to invoice line. A header chantier field alone does not allocate journal lines.

Choose a single cost recognition policy

Event	Management interpretation
Approved purchase order	Commitment, not actual consumed cost
Receipt of materials	Inventory value, subject to valuation configuration
Transfer to internal site stock	Inventory relocation, not new cost
Material consumption	Site cost according to approved consumption policy
Posted subcontractor bill	Direct service cost
Posted employee expense	Site expense if allocated
Posted customer invoice	Invoiced revenue, net of relevant reversals

Supplier material bills and consumed materials must not both be counted as two independent site costs. Determine which accounting/analytic entries represent the material cost and exclude/reclassify the other representation in the report. Invoiced revenue is not necessarily earned revenue; label it accurately.

Optional labor and profitability

If timesheets are used for labor estimates, distinguish estimated operational cost from posted payroll cost. Do not add both for the same labor without an explicit adjustment/reconciliation policy. Profitability can show budget, commitments, recognized costs and invoiced revenue in separate columns.

Acceptance gate

Receive 100 units, transfer 40 to site, consume 25 and return 15. The site has zero remaining units; main stock has 75. Only 25 consumed units contribute material cost under the consumption policy. Reversing a consumption must reverse its corresponding cost attribution once, retaining the audit trail.

08 | Reconciliation, reminders and cheques

Bank import and matching

1. Select a verified Odoo 17 Community-compatible statement import/reconciliation solution. Confirm supported CSV/OFX formats with real anonymized bank samples.
2. Map date, amount, reference, partner and bank account fields. Define debit/credit sign conventions and statement opening/closing balances.
3. Define an import identity/deduplication strategy. Reimporting the same statement must not create another set of transactions.
4. Configure exact-reference and amount/partner matching where reliable. Handle bank fees, partial payments, grouped payments and unknown counterparties explicitly.
5. Present ambiguous candidates for accountant review. Match to existing payments/entries before generating any new accounting effect.
6. Reconcile and verify statement balances against the bank journal.

The report's AI requirement is separate from deterministic rules. Define expected benefit, sample cases, confidence thresholds and review behavior. Select and validate a matching addon/service or obtain a documented scope change. Do not label rules as AI or permit an unverified suggestion to post accounting entries unchecked.

Overdue follow-up workflow

1. Configure templates for 7, 15 and 30 days after the due date, using the customer's language and outstanding balance.
2. Add dispute/hold status and a responsible collector. Define whether aging follows individual installment due dates.
3. Schedule a daily company-aware job. Select posted customer receivables with residual balances and eligible due dates.
4. Recheck payment and hold status immediately before sending. Log invoice, level, recipient and send result.
5. Use an idempotency key so retries do not send the same reminder twice. Escalate undeliverable messages to an activity.
6. Stop further reminders when paid or held. Define a controlled release procedure for disputes.

Cheque printing

Use a compatible payment/cheque foundation, then implement the actual bank layout in QWeb where needed. Configure cheque numbering, payee, words/amount formatting, date and page alignment. Test on blank sample paper. Restrict print/void/reprint actions and retain the original number and reason. A printed cheque is not proof of bank clearance.

Acceptance gate

Reimport a bank sample without duplication; reconcile one partial payment and one fee; leave one ambiguous payment unposted for review. Confirm paid and disputed invoices receive no reminder. Validate cheque alignment, numbering and void/reprint history with the accountant.

09 | CRM configuration and daily workflow

Configure the sales pipeline

1. Install CRM. In its settings, enable Leads if a qualification step is needed before opportunity creation.
2. Create sales teams and assign members. Set ownership rules for private, team-shared or company-wide opportunities.
3. Create stages: New, Qualified, Proposal, Negotiation and Won. Use the Lost action and a controlled list of lost reasons for unsuccessful deals.
4. Add qualification fields: work type, site address, region, approximate contract value, expected start, decision maker and next action.
5. Create activity types for calls, site visits, quotation preparation and follow-up. Define responsible users and default deadlines.
6. Define team targets by month and currency. Decide whether attainment is based on confirmed orders or invoiced revenue; report both separately if useful.
7. Configure email routing so replies are attached to the right lead/opportunity. Test external sender, reply and attachment behavior.

Lead-to-order operating procedure

1. Salesperson records the lead and contact details, including source and proposed works.
2. Check for an existing contact/opportunity before creating another record.
3. Qualify the need, expected date and decision maker; schedule a site visit.
4. Convert to an opportunity and assign the correct customer. Record visit findings in chatter and attach scope documents.
5. Create the quotation from the opportunity; include the proposed chantier or site reference.
6. Send the quotation and move to Proposal. The actual send event starts the J+3 follow-up rule.
7. Complete follow-up and record the outcome. Move to Negotiation when commercial terms are being revised.
8. On accepted order, mark won according to the agreed process. Otherwise use Lost with a reason; retain history.

J+3 automation specification

Use a Python addon or a verified automation mechanism. Store the quotation send timestamp and an identifiable follow-up activity. Create one activity per defined quotation cycle, assigned to the salesperson. Suppress it when the order is already confirmed/cancelled or the deal is closed. Define whether resending an unchanged quotation resets the clock; default to no duplicate activity. An email failure should not be treated as successful customer delivery.

Acceptance gate

Capture one lead, convert it, create/send a quotation and complete the follow-up. Run the automation twice and verify only one eligible activity exists. Confirm one lost reason and one won opportunity appear correctly in CRM reporting. Mailbox synchronization beyond routing/logging remains a separately tested integration requirement.

10 | Sales configuration and order processing

Configuration actions

1. Install Sales and inventory integration as required. Add project integration only after deciding how service orders create or link projects/tasks.
2. Enable pricelists. Create Individual, Developer and Business lists; configure quantity breaks and date-limited rules where required. Assign the intended list to each customer.
3. For regional pricing, map customers/regions to pricelists or implement a validated custom rule. Do not assume a region field automatically changes pricing.
4. Create quotation templates for construction, carpentry and finishing, with scope, exclusions, payment terms and validity period.
5. Configure materials for the agreed ordered/delivered invoicing policy. Configure service billing separately. Confirm each product's route and taxes.
6. Enable customer portal and online quotation signature. Configure who is allowed to accept and what records are visible.
7. Customize QWeb quotation/order layouts with company details, work sections, quantities, unit prices, totals and terms. Test multi-page line tables.
8. Add a chantier link with company validation. Permit selecting an existing project for additional orders; do not create another site for every quotation.

Salesperson procedure

1. Create a quotation from the opportunity or customer. Confirm delivery/invoice addresses and pricelist.
2. Add measured quantities and service scope. Review taxes and expected dates.
3. Obtain internal review where margins, discounts or commitments require approval.
4. Send the quotation through the intended channel; customer views and signs through the portal.
5. Confirm the order once and initialize/link the chantier using an idempotent action.
6. Generate deliveries for sold stock items. For a construction service, initiate the separate material-request workflow instead of assuming service lines consume materials.
7. Record fulfilled quantities or approved service progress. Create a draft invoice from eligible order lines.
8. Accountant checks/posts the invoice; payment collection follows the finance workflow.

Exceptions

Revised quotations retain history and clear version references. Additional scope creates an approved variation/additional order according to policy. Cancellation must handle related draft invoices and pending deliveries without deleting completed stock or posted accounting records. Customer returns use stock return operations and a corresponding credit-note decision.

Acceptance gate

Portal users see only their authorized documents. Acceptance produces one order and one intended chantier link. Test volume pricing, a partial delivery, partial invoicing and cancellation. The invoice retains its order-line/chantier origin and correct analytic allocation.

11 | Construction billing and optional extensions

Select the contractual billing method

Method	Implementation
Material sale	Standard ordered/delivered quantity policy
Fixed-price service	Standard sales invoice process with agreed fulfillment control
Advance payment	Standard down-payment invoice, accountant-approved accounts
Certified progress	Custom approved progress records if required
Retention / variation	Separate specification and accounting treatment

Down payment and final billing

1. Confirm the contract/order and record payment terms.
2. Use Create Invoice to prepare the agreed percentage/fixed advance. Verify the amount, tax treatment and accounts with the accountant.
3. Post/send the advance invoice and record/reconcile the payment when received.
4. Perform work and record fulfillment under the chosen contract method.
5. Prepare the subsequent/final invoice with the applicable down-payment deduction using Odoo's supported process.
6. Review the cumulative invoiced amount and residual customer balance. A down payment must not be deducted twice.

Optional certified progress addon

Use a progress certificate linked to the sales order and chantier, with lines linked to contract items. Store contract quantity/price, prior certified quantity, current accepted quantity, cumulative quantity and supporting documents. Suggested states: Draft → Submitted → Reviewed → Approved → Invoiced; rejection returns to a controlled correction stage.

1. Chantier manager enters quantities actually completed and attaches measurement evidence.
2. Authorized reviewer checks quality/completion and contractual limits. Customer acceptance is recorded if required.
3. Approver freezes the approved certificate version. Increased scope needs an approved variation rather than a silent contract quantity change.
4. Create a draft invoice only for approved, unbilled progress. Link generated lines back to the certificate.
5. Prevent a second invoice generation for the same certified portion. Handle corrections through explicit reversal/reissue actions.

If retention is required, define its basis, percentage, due date, release conditions and balance-sheet treatment with the accountant. Do not implement it merely as an unexplained discount. Revenue recognition, advances and invoicing may differ; report them separately where necessary.

Acceptance gate

For a contract with an advance and two progress invoices, cumulative billing never exceeds approved contract/variation value and the advance is deducted exactly as agreed. A rejected/unapproved certificate cannot generate an invoice. These construction additions remain optional until confirmed; they are not explicitly specified in the source report.

12 | Purchase configuration and requisitions

Configuration actions

1. Install Purchase and purchase-stock integration. Verify whether purchase agreements are enabled through `purchase_requisition` in the selected installation.
2. Import approved suppliers, lead times, negotiated prices, purchase units and payment terms.
3. Enable order approval and configure the report's 10,000 MAD threshold. Record tax basis, equality behavior and foreign-currency conversion policy.
4. Assign buyer and approver roles separately. Native manager rights may permit confirmation: add Python checks if two different people are mandatory.
5. Configure received-quantity billing for relevant materials. For services, define how the receiver confirms service completion.
6. Define permissible price/quantity differences and exception approvers. Log reasons rather than silently changing matched records.
7. Configure recurring supplier agreements and validity dates. Check actual applicability before adopting an agreement price.

Material request data

Create a request header with company, chantier, requester, required date, priority and justification. Lines contain product, unit, requested quantity, available/reserved quantity and suggested fulfillment source. Store links to transfer moves and purchase lines; support partial fulfillment.

Request-to-order workflow

1. Chantier manager submits the need with quantity and required date.
2. Storekeeper checks available stock and existing incoming supplies. Reserve eligible on-hand quantities.
3. Split fulfillment: transfer available material; procure only the shortage. Do not count quarantined stock as available.
4. Buyer creates RFQs from shortages and compares price, delivery, payment terms and suitability.
5. Select the vendor and record selection rationale when policy requires comparison.
6. Submit the purchase order for approval. Approver reviews total, chantier, budget context and delivery need.
7. Confirm after approval. Changes to critical fields after approval must trigger reapproval under the chosen policy.
8. Receive/inspect; create the bill from the order and resolve mismatches before posting/payment.

Acceptance gate

Test totals of 9,999, 10,000 and 10,001 MAD under the signed threshold rule. Test buyer self-approval, another company's approver and a post-approval price increase. One request fulfilled partly from stock and partly from purchase must show accurate outstanding quantities without duplicate procurement.

13 | Mandatory receiving quality workflow

Standard foundation versus required enforcement

Use standard receipt routes and stock movements, plus an extension for inspection records and release controls. A multi-step warehouse receipt can move through Input → Quality → Stock. These internal locations represent physical custody; financial valuation timing follows configured Odoo valuation rules and must be verified, not assumed to wait for inspection.

Configure

1. Enable storage locations and multi-step routes in Inventory settings. Configure the warehouse's receipt steps.
2. Create Input, Quality and a separate Quarantine/Rejected location. Restrict ordinary picking from these locations.
3. Define inspection templates per material/product category: ordered quantity, condition, dimensions/specification, lot/serial, certificates and photos.
4. Create an inspection model linked to receipt, stock moves and relevant lot/serial records. Store accepted/rejected quantities, inspector, date and reason.
5. Generate inspections when eligible receipts are processed. Define whether inspection is per line, lot or sample.
6. Block Quality → Stock release unless the specific quantities/lots are approved. Enforce on the server, including imports, alternate forms and automatic transfer paths.
7. Permit only designated quality users to approve. Reinspection or override requires reason and history.

Storekeeper and inspector procedure

1. Open the expected receipt and compare supplier delivery documents with the order.
2. Enter actual received quantities and lot/serial identifiers. Validate only physically received quantities; create a backorder for the rest if expected.
3. Move received material into Quality and open its inspection.
4. Inspector records results, evidence and pass/fail quantities.
5. Release approved quantities to usable stock using the linked transfer.
6. Move rejected quantities to Quarantine. Buyer decides return/replacement with the supplier.
7. Use a stock return referencing the original receipt when returning goods. Accountant separately decides supplier credit/bill adjustments.

Acceptance gate

Receive 10 units: approve 8, reject 2. Only 8 become available for ordinary site requests. Attempt to release the rejected lot, validate a direct bypass transfer or reuse the same approval for additional quantities: all must fail. Confirm backorders and return quantities remain linked to the correct purchase line. Do not confuse physical receipt with quality acceptance in purchasing reports.

14 | Inventory setup and replenishment

Configuration actions

1. Enable storage locations, multi-step routes, lots and serial numbers as required. Create main and site internal locations under the correct company.
2. Set material categories to AVCO where the report calls for weighted average cost. Choose FIFO only for categories explicitly approved for it; one category has one configured costing method.
3. Configure automated valuation and related accounts before opening stock. Test receipts, outflows, price differences and returns with the accountant.
4. Configure units of measure, traceability and vendor lead times on products. Unique equipment should carry serial numbers.
5. Define operation types for receipts, internal transfers, site consumption, returns and counts as the approved design requires.
6. Select a compatible scanning interface. Test product, lot/serial and location barcodes with the actual device; labels alone do not deliver guided barcode operations.

Replenishment procedure

1. Open a storable product's Reordering Rules (or the replenishment configuration view).
2. Choose the target location, min/max quantity, order multiple and Buy route. Ensure the product has a valid vendor and purchasing data.
3. Decide manual review versus automatic procurement. Start with manual review until forecasts are trustworthy.
4. Trigger a demand that reduces forecast stock below the minimum. Review generated replenishment quantities, including incoming supplies.
5. Run/allow the standard scheduler as configured. Verify creation of the expected RFQ; this must not bypass purchase approval.
6. For chantier supply from the main warehouse, configure internal resupply consistently. Do not let a site independently buy material that is already reserved at the main warehouse.

Counts and equipment

Assign counts by location and responsible person. Record physical counts, investigate differences and validate adjustments under the approved role policy. Retain reasons. For serialized equipment, track issue, site-to-site transfer and return. A reusable tool issued to site is not consumed material; capitalization/depreciation is a separate accounting decision.

Acceptance gate

Lower forecast stock below minimum and confirm correct RFQ quantity and vendor. Test order multiples and an existing incoming order to avoid overbuying. Scan a product and serial correctly; reject an invalid/duplicate serial as appropriate. Reconcile stock quantities and valuation after a receipt, transfer, count adjustment and supplier return.

15 | Site transfers, consumption and returns

Design the consumption extension carefully

Standard internal transfers relocate company-owned stock. They should not by themselves create site material expense. Define the intended consumption operation, destination semantics, accounting effect and analytic attribution with the accountant before coding. Do not use an inventory-loss location as a generic consumption shortcut.

The addon should reuse stock moves and valuation layers; it must not independently invent a second valuation engine. Link every attributed cost to a source movement/valuation event. Address subsequent valuation adjustments and reversals. Where standard accounting does not provide the needed analytic attribution, add a controlled bridge with unique origin references.

Daily workflow

1. Chantier manager creates/submits a material request with the required date and work package.
2. Storekeeper checks and reserves available stock. Approver authorizes the request if policy requires it.
3. Create the internal transfer Main Stock → Chantier Stock. Record exact lot/serial quantities and dispatch evidence.
4. Confirm actual site receipt. If separate dispatch and arrival evidence is required, design a transit/receipt process rather than claiming one validation proves both.
5. Manager records actual consumption by product, work type, date and quantity, referencing available site stock.
6. Authorized user approves/validates consumption. The addon creates or links the stock effect and cost attribution once.
7. Return unused materials through an internal transfer to Main Stock. Returned unused material is not a reversal of consumption because it was never consumed.
8. Correct erroneous consumption through a linked reversal. Do not delete completed moves or their analytic history.

Controls and closure

Restrict consumption to open authorized sites and eligible approved stock. If the business prohibits negative stock, implement and test that rule; standard workflows must not be assumed to enforce it universally. Concurrency checks are needed when two users consume the same available quantity. At closure, reconcile unused material, equipment, pending transfers and unresolved adjustments.

Acceptance gate

Using 100 units initially in main stock: transfer 40, consume 25, return 15. Final main stock is 75; site stock is zero; consumed quantity is 25. At an unchanged cost of 50 MAD/unit, attributed material cost is 1,250 MAD. This example excludes taxes, other receipts and valuation adjustments. Retrying the action must not double the 1,250 MAD attribution.

16 | HR configuration and confidentiality

Standard and extension scope

Install `hr`, `hr_contract`, `hr_holidays` and `hr_recruitment`. Add attendance/timesheets only if required. Appraisals need a suitable Community extension or custom process. Full Moroccan payroll is not specified by the report; do not claim employee/contract configuration delivers payroll compliance.

Configuration sequence

1. Create Direction, Administration, Commercial, Technical and Chantier departments, then job positions and reporting lines.
2. Import employees with professional contact details. Restrict private addresses, identification, bank details and contracts to authorized HR users.
3. Create contract categories for CDI, CDD, temporary and seasonal arrangements. Store dates, working schedule and supporting documents.
4. Configure working schedules and public holidays validated by HR. The report's 44-hour week is an input for review, not a universal hardcoded rule.
5. Create annual, sickness and family-event leave types. Set approval, supporting evidence, allocation and accrual policies.
6. Validate the report's annual-leave assumption against actual eligibility and accrual rules. Configure these as data, not fixed Python constants.
7. Assign employee managers and leave approvers. Define substitute approvers for absence.
8. Configure recruitment stages: Application, Screening, Interview, Offer, Hired. Set ownership and candidate privacy.
9. Define appraisal frequency, objective categories, employee self-assessment and manager review rights.
10. Schedule contract/document expiry activities. Keep sensitive attachments inside the appropriate HR permissions.

Optional site time and expenses

If hours are required, link time entries to the correct project/task and define who validates them. Separate attendance (presence) from productive/billable time. If expense reimbursement is required, use `hr_expense`: employee attaches receipt and allocation → manager reviews → accountant posts → reimbursement follows payment controls.

Acceptance gate

Employee can create an own leave request but cannot change an approved allocation. A manager sees their team's operational availability without unrestricted access to every confidential record. A storekeeper cannot download an employee contract through a direct attachment URL. An HR expiry activity reaches the assigned owner at the intended lead time.

17 | HR operating workflows

Leave request to availability

1. Employee selects leave type and dates in Time Off. Verify the system calculates working days/hours from the assigned schedule.
2. Attach required evidence and submit. The request becomes noneditable except through the approved correction path.
3. Manager reviews site/team coverage and either approves or refuses with a reason.
4. HR performs the second validation if configured, checking entitlement and documentation.
5. Approved leave updates availability and the dashboard source. Keep medical details confidential.
6. A cancellation follows the configured approval path and restores allocation correctly. Test public holidays, weekends and overlapping requests.

Recruitment to employee record

1. Create a job position and responsible recruiter.
2. Capture candidate details and application documents; check for duplicates.
3. Move through screening/interview stages and schedule interviews in Calendar.
4. Record assessments with restricted access. Set a next activity for every active candidate.
5. Send/record the offer through the approved business process.
6. On acceptance, create the employee from the candidate, verify transferred fields and create the contract.
7. Close/archive the application according to retention policy. Do not make all candidate assessments visible in the general employee directory.

Periodic appraisal

1. HR opens a review for the employee, manager and review period, with objectives and due dates.
2. Employee submits self-assessment. Manager reviews performance and supporting evidence.
3. Schedule the review meeting. Record agreed outcomes, development actions and next review date.
4. Approve/close the review. Completed content is protected from casual edits; an authorized reopening records the reason.
5. Create activities for agreed actions and monitor completion without exposing confidential evaluations to general project users.

Optional expense workflow

Employee creates expense with receipt, amount, payment mode and chantier distribution → manager validates business purpose → accountant verifies accounts/taxes and posts → authorized payer processes reimbursement → reconciliation closes the balance. Reject missing evidence or unsupported allocation before posting.

Acceptance gate

Run one approval, one refusal and one cancellation of leave. Confirm balance and daily absence KPI. Convert one candidate without duplicate employee records. Close an appraisal and test denied access from another employee. If expenses are enabled, ensure a rejected expense never appears as a posted site cost.

18 | Central document management

Select the GED foundation

Standard business attachments are useful but do not provide the whole centralized document-management requirement. Select a maintained 17.0-compatible document addon that supports the needed workspace, metadata, search, versioning and access behavior. Verify these capabilities in a trial rather than assuming every GED addon provides them.

Configure step by step

1. Create workspaces for Commercial, Purchasing, Chantiers, Accounting and HR. Define company ownership and restricted subfolders.
2. Define document categories: contract, quotation, invoice, delivery note, drawing, inspection, certificate and employee document.
3. Define required metadata by category: company, linked business record, chantier, owner, issue date, expiry date and revision.
4. Assign workspace access based on business roles. Define what happens when a linked record becomes inaccessible or is archived.
5. Choose the versioning rule: new revision preserves previous content, author, date and approval status. Avoid overwriting an approved technical drawing without history.
6. Define Draft → In review → Approved → Superseded/Archived for documents that need approval. Ordinary invoice attachments need not all use this lifecycle.
7. Configure a bridge that classifies generated business documents and links existing attachments without uncontrolled duplicate copies.
8. Configure expiry activities for expiring contracts/certificates, assigned to a responsible owner.

Daily procedure

1. Upload or generate the document from its originating business record.
2. Verify category, company, project and owner; fill missing metadata.
3. Submit for review when required. Reviewer checks content and approves/rejects with a comment.
4. Make the current approved version visible to authorized teams.
5. For revision, upload a new version and reapprove if required; retain earlier approved versions.
6. Archive obsolete content according to the agreed retention policy. Archiving a project must not destroy its accounting evidence.

Portal and privacy

Grant external access only to explicitly allowed documents. Do not expose an entire chantier workspace merely because a customer can view its quotation. Test attachment/download authorization directly, including links copied from another user's session. HR documents require stronger restrictions than general project drawings.

Acceptance gate

Find all authorized documents for one chantier by category. Replace a drawing with revision 2 and retrieve revision 1. Verify a general employee cannot retrieve private contracts, and another company's user cannot access restricted files. Expiry automation creates one actionable reminder, not repeated duplicates.

19 | Discussion, chatter and email

Configure the communication layer

1. Use mail/Discuss with business record chatter and activities. Configure the company's Odoo outgoing/incoming email behavior using its chosen provider; infrastructure setup remains outside this guide.
2. Create channels for management, administration and operational teams. Define membership and who can invite others.
3. Prefer record chatter for transaction decisions so messages remain linked to orders, projects and invoices.
4. Define templates for quotations, purchase communications, approvals, payment reminders and document expiry. Set recipient rules and language selection.
5. Review automatic followers. A contact following a record may receive notifications; do not subscribe external parties indiscriminately.
6. Train users on Send message versus Log note. Internal discussion must use the appropriate internal channel/note behavior.
7. Define incoming aliases where needed and test routing to existing records versus creation of new leads.

Daily workflow

1. Open the business record before discussing a request or exception.
2. Add an internal note for colleagues; attach evidence and mention only the relevant people.
3. Create an activity with owner, due date and requested outcome when someone must act.
4. Owner performs the action and records completion or a documented exception.
5. Send external communication from the correct record/template after checking recipients and attachments.
6. Monitor delivery failures and incoming replies. Assign an activity to resolve failed or unlinked messages.

Integration boundaries

Routing incoming messages, logging outgoing messages, full mailbox synchronization and calendar synchronization are different features. Write down which is required for Gmail/Outlook. Validate the installed connector's behavior; do not promise full mailbox mirroring from basic Odoo email configuration.

Automated communications

Configure domain-specific triggers with idempotency, status rechecks and company-aware recipient selection. Avoid sending a quotation follow-up for a confirmed order or a reminder for a paid invoice. Keep a visible history of automated actions and delivery errors.

Acceptance gate

Send a test quotation, reply externally and confirm the reply attaches to the intended record. An internal note must not arrive in the customer's mailbox. Test a failed address and ensure an owner can see and resolve the failure. Verify channel membership and followers do not leak HR or financial information.

20 | Calendar and business appointments

Configure

1. Install/use Calendar and confirm each user's Africa/Casablanca time zone. Test a date affected by local clock changes where applicable; do not substitute a fixed UTC offset.
2. Define appointment categories for site visits, supplier meetings, recruitment interviews and appraisals.
3. Set default reminders, event privacy and attendee rules.
4. Confirm how events link to contacts, opportunities and other source records. Add chantier or HR links through extensions only where standard links are insufficient.
5. If Google/Microsoft synchronization is required, select and configure the relevant integration and verify its supported direction and conflict behavior.
6. Define which business events create appointments, which create activities and which remain visible in module scheduling views.

Appointment workflow

1. From the relevant lead/project/HR process, prepare an event with subject, date, duration, location and owner.
2. Add only the necessary participants. Check confidentiality before adding external attendees.
3. Send the invitation and review acceptance/refusal responses.
4. Update or reschedule through the agreed source system. Confirm the change reaches attendees without duplicate events.
5. After the meeting, record the result on the originating business record rather than leaving important decisions only in the calendar description.
6. Create follow-up activities with owners and deadlines.

Define the report's 'all modules' expectation

Process	Recommended scheduling behavior
CRM site visit	Calendar event linked to opportunity/customer
Purchase follow-up	Assigned activity; meeting only if necessary
Stock delivery	Stock scheduled date and delivery report
Overdue invoice	Collection activity/reminder workflow
Recruitment/appraisal	Restricted calendar meeting and HR source record
Chantier coordination	Calendar event linked to project where implemented

Do not turn every transaction into an event automatically. Define measurable coverage for the required cross-module scheduling instead.

Acceptance gate

Create, reschedule and cancel one internal and one external appointment. Verify time zones, reminders and linked records. Where synchronization is selected, test edits in both supported directions and duplicate prevention. A non-HR user must not see confidential appraisal content merely because the meeting blocks a colleague's time.

21 | Management KPI definitions

Implement the eight report KPIs

KPI	Exact baseline definition and source
Monthly revenue	Posted entries on approved revenue accounts for the month, with correct sign; excludes tax and nets relevant credits. Source: account.move.line.
Invoices unpaid >30 days	Outstanding customer receivable amounts with due dates more than 30 days before the as-of date. Respect installments and residuals.
Quotation conversion	Confirmed quotation/order count divided by eligible sent quotation count for the same defined cohort. Explicitly handle revisions and cancellations.
Current stock value	Company inventory valuation at the report date using the configured valuation source; not current quantity multiplied by today's list price.
Pending supplier orders	Confirmed POs with remaining receipt obligations. Show awaiting-approval orders as a separate measure.
Pipeline opportunities	Open opportunities by stage, count and expected value. Distinguish unweighted value from probability-weighted value.
Today's absences	Approved time off overlapping the local working day. Count distinct employees; attendance exceptions are separate.
This week's deliveries	Pending outgoing deliveries scheduled in the defined local week. Internal site transfers appear separately unless explicitly included.

Filters and presentation

Use company and date/as-of filters on every applicable KPI. Add team, chantier and region filters only where data supports them. A current pipeline is a snapshot, whereas monthly revenue is a period flow: label these differences. Historical aging or pipeline requires a true historical computation/snapshot; present-day residuals cannot reliably reconstruct every past position.

Consolidated views

Management may aggregate authorized companies, but mixed currencies require a defined reporting currency and exchange-rate policy. Basic multi-company totals are not statutory consolidation. Intercompany eliminations and formal consolidated statements require a separate accounting specification.

Optional chantier indicators

Show committed spend, consumed material cost, posted direct service expenses, invoiced revenue and remaining site stock in separate measures. Add margin only after the cost and revenue bases have been approved. Do not mix tax-inclusive and tax-exclusive amounts.

Acceptance gate

Each KPI has an owner, definition, source domain, sign/date policy, refresh behavior and drill-down. The same filters produce matching totals in the underlying module report. For a revised quotation, partial payment, credit note and partial receipt, verify the KPI follows the written definition.

22 | Dashboard build and validation

Development steps

1. Start with saved list/pivot/graph views to validate source logic. Do not build attractive cards before agreeing on the calculation.
2. Implement an elmokrif_dashboard addon using standard actions/views where sufficient and an OWL client action only for the combined dashboard where needed.
3. Provide server-side methods that accept validated filter inputs. Use ORM aggregation and existing accounting/valuation reports where practical.
4. Apply user permissions and company context to queries. Avoid unrestricted sudo or raw SQL that bypasses record rules and exposes unauthorized totals.
5. Add drill-down actions with exactly the domain used by the displayed figure. Show the active filters and reporting basis.
6. For expensive snapshots/caches, include company, date, permission scope and other relevant filters in cache design. Invalidate when source changes or display an accurate refresh timestamp.
7. Define daily refresh for absence/aging/delivery summaries if selected; use live or appropriately refreshed values for the other metrics. 'Real time' must reflect actual behavior.
8. Handle empty datasets and unavailable permissions with clear messages, not misleading zeros.

Reconciliation procedure

1. Select one company and a short test period with known records.
2. Export or inspect the source report for each KPI.
3. Compare count/amount and investigate rounding, dates, residuals, returns, credit notes and cancelled records.
4. Click the KPI and confirm the same source records appear.
5. Repeat with a user restricted to one team/site and with an authorized multi-company user.
6. Post a transaction and check the expected refresh. Archive a user/project and confirm history remains meaningful.

Performance and application behavior

The source report asks for sub-two-second response for 20 concurrent users. Treat this as an application acceptance target to measure on representative data, including custom dashboard queries. Optimize expensive aggregations and avoid reading every transaction into Python. Infrastructure tuning is outside this guide, but query design and observed response time belong to application validation.

Acceptance gate

Management signs off the eight numbers and their drill-downs. Record representative response times and the dataset size. There must be no aggregate information leak across companies or HR permissions. Do not close the dashboard requirement while accounting/stock sources remain unreconciled.

23 | Custom addon boundaries

Proposed modules and responsibilities

Addon	Depends on / extends	Deliverable
elmokrif_chantier	project, analytic, stock, mail	Site master, lifecycle, account/location initialization
elmokrif_sale_chantier	sale_management, sale_stock, chantier	Sales links, analytic propagation, project creation rules
elmokrif_purchase_chantier	purchase_stock, chantier	Requests, stock/purchase split, approval controls
elmokrif_stock_chantier	stock_account, chantier	Consumption/returns and source-linked cost bridge
elmokrif_quality	stock, purchase_stock, mail	Inspection templates/results and release enforcement
elmokrif_account_extension	account, analytic, selected finance addons	Missing controls, follow-ups, local document adaptations
elmokrif_hr_extension	Relevant hr modules	Appraisal/expiry gaps and optional site assignment
elmokrif_documents_bridge	Selected GED and chantier	Classification, links and controlled automation
elmokrif_dashboard	Required operational modules	Eight KPIs, filtering and drill-downs

Keep optional certified-progress billing and expense/time integration in separate bridge addons if they introduce extra dependencies. Do not force HR or Documents into the chantier core merely for future possibilities.

New records worth implementing

Material request/header-lines; quality inspection/template/result lines; approved consumption record/lines if standard movements alone do not capture approvals; optional progress certificate/lines; appraisal records if not supplied by the selected extension. Reuse standard orders, invoices, stock moves and valuation layers for actual transactions.

Dependency strategy

Core provides project references and business roles. Domain bridges add links to their own standard modules. Cross-domain automation belongs in the narrow bridge that owns the transaction. The dashboard consumes business data and must not be required for posting invoices or validating stock.

Existing customization compatibility

If a chantier model already exists in your database, inventory its fields, records and links before adopting project.project. Choose one authoritative master. Either migrate existing records with an explicit mapping or retain the existing model with a controlled one-to-one project relationship. Do not create two competing site masters or delete historical links.

Acceptance gate

Each module installs/upgrades with declared dependencies and no circular imports. Removing an optional dashboard/bridge dependency must not compromise unrelated accounting history. Every custom model has ownership, company rules, source links and a documented lifecycle.

24 | Implementation rules for Python addons

Minimum module contents

Use `__init__.py`, `__manifest__.py`, models, security and views; add data for sequences/jobs, report for QWeb outputs and tests for meaningful business invariants. The manifest declares dependencies and loads security/data/views in a valid order. Prefix custom external IDs consistently.

Build each feature in this sequence

1. Define business fields, required data and state transitions before writing views. Decide who can create, submit, approve, cancel and reopen.
2. Implement models and constraints: positive quantities, valid companies, compatible units, source references and forbidden transitions.
3. Implement approval actions with server-side group/role checks. Preserve Odoo super behavior when extending methods; do not replace a large standard method unnecessarily.
4. Add access CSV and record rules. Test both authorized and forbidden cases, including direct method calls/import paths.
5. Build list/form/search views and smart buttons. For Odoo 17, use supported direct invisible/readonly expressions rather than copying older `attrs/states` view syntax.
6. Add activities, sequences and templates only where required. Keep threshold and timing parameters configurable by authorized users.
7. Implement transaction creation with unique origin links and duplicate prevention. Use robust constraints/locking where concurrent approvals could create two effects.
8. Add cancellation/reversal paths and audit messages. Never delete posted entries or completed stock movements to 'undo' business history.
9. Test install and upgrade plus the affected end-to-end workflow. Retain representative evidence and version notes.

Field and security decisions

Use Many2one relationships with company checks where suitable. Propagate chantier at line level when a document can cover several projects. Header defaults are convenient but cannot override line allocations. Ensure smart-button counts and related fields do not expose records hidden from the user.

Financial integrity

Link analytic effects to standard financial/valuation origins. Enforce one effect per source, reversal and adjustment identity. Reconcile after price changes, returns, credits and valuation adjustments. Do not compute historical stock cost from today's product cost field.

Acceptance gate

Tests cover multi-company isolation, threshold boundaries, duplicate action/retry, concurrent consumption risk, missing analytic allocation, quality bypass and reversal effects. View rendering alone is insufficient. Tests should validate business outcomes rather than simply mirror the implementation.

25 | Automation and approval specifications

Required automated behavior

Trigger	Action	Stop / duplicate rule
Quotation sent	J+3 salesperson activity	No duplicate eligible cycle; stop if confirmed/cancelled
Forecast below minimum	Standard replenishment proposal/RFO	Respect incoming supply and purchase approval
Receipt requiring inspection	Create inspection	One defined inspection scope per receipt/lot
Approved site consumption	Stock/cost effect	Unique source; never repeat on retry
Invoice overdue	J+7/J+15/J+30 follow-up	Residual >0, no hold, one send per level
Contract/document nearing expiry	Assigned activity	One outstanding action per expiry event
Appraisal due	Review/activity creation	One review per employee/period

Common job design

1. Define the exact event, eligible records, company context, responsible user and deadline basis.
2. Store a durable origin identifier and processing status. An activity's text alone is not a reliable duplicate key.
3. Recheck eligibility just before mutation or sending; another user may have paid, cancelled or approved the record.
4. Process bounded batches and record failures visibly. A single malformed record should not silently halt all future reminders.
5. Use a restricted service context appropriate to the action. Avoid broad elevated access that bypasses business scope.
6. Separate retryable delivery errors from successful sending. Retain sufficient evidence to investigate and safely retry.

Approval evidence

Store requester, submission time, approver, approval time, approved value/version and reason for rejection/override. Define which changes invalidate approval: product, quantity, price, company, chantier or planned commitment. A 'manager' group should not automatically override a requirement for two distinct people.

Configuration versus code

Set timing, thresholds, templates and approver mappings as configuration data where practical. Put critical security, state transitions and financial integrity in Python. Do not depend on UI-only mandatory fields or a scheduled job to enforce rules that must hold immediately.

Acceptance gate

Run each job twice against the same dataset; no duplicate business effects or communications should appear. Change the source state immediately before a retry and verify suppression. Confirm failure visibility and authorized recovery. Test the report's approval threshold at equality and in another currency under the documented conversion rule.

26 | Pilot A: purchase to site consumption

Test setup

Create test-only data: chantier CH-TEST-001, one buyer, another purchase approver, one storekeeper and a quality inspector. Use a stock-managed material with unit U, AVCO, a validated receipt route and no opening quantity. Price is 50 MAD/U. All amounts below exclude taxes and other valuation effects; this is a mechanical test, not a tax/accounting policy example.

Execute with the actual role accounts

1. Chantier manager requests 240 U. Available stock is zero, so the buyer prepares an RFQ for 240 U at 50 MAD: 12,000 MAD before tax.
2. Submit the PO. Confirm the buyer cannot self-approve under the selected two-person policy. The designated approver reviews and approves it.
3. Receive 100 U of the 240 U into Input; create a backorder for 140 U if still expected. Record any required lot.
4. Inspect the received 100 U: pass 90 U, reject 10 U. Attempt to release all 100 U and confirm the quality control prevents it.
5. Release 90 U to Main Stock. Hold 10 U in Quarantine, then perform a supplier return linked to the receipt. Review the purchase/request outstanding quantities after return; decide whether replacement is expected.
6. From approved Main Stock, transfer 40 U to CH-TEST-001. Confirm the request's fulfilled and remaining quantities update correctly.
7. Record consumption of 25 U with work type and date. Validate once; retry the action to prove no duplicate stock/cost effect.
8. Return 15 unused U to Main Stock. Keep the 140 U backorder and any replacement decision visible; do not close the request just because one transfer completed.
9. Create the vendor bill from the PO for the agreed billable quantity after resolving rejection/return. The accountant must review Odoo's received/billable quantities; physical receipt is not identical to accepted quantity.

Expected physical result after supplier return

Main Stock: 65 U (90 accepted minus 40 issued plus 15 returned). Site Stock: 0 U. Consumed: 25 U. Returned to supplier: 10 U. Total net company receipt: 90 U. If cost remains 50 MAD/U, remaining inventory is 3,250 MAD and consumed material cost is 1,250 MAD under the approved design.

Evidence to retain

Request, PO approval history, receipt/backorder, inspection, release transfer, supplier return, site transfer, consumption, internal return and reviewed bill. Reconcile physical quantities and valuation separately. Any variance must be explained before moving to the sales/finance pilot.

27 | Pilot B: sales, billing and collection

Test setup

Create one test customer, opportunity and 30,000 MAD service contract linked to CH-TEST-001. Use the accountant-approved invoice/tax configuration. Figures below show the simplified tax-exclusive commercial amounts; actual receivable/reconciliation amounts include applicable tax and rounding.

Execute

1. Salesperson captures and qualifies the opportunity, schedules a site visit and creates a 30,000 MAD quotation.
2. Send it through the portal. Run the J+3 automation twice and verify a single eligible follow-up activity.
3. Customer accepts/signs; confirm the order and link the existing chantier. Repeat the initialization action and verify no duplicate site/account/location.
4. Prepare a 20% down-payment invoice: 6,000 MAD before applicable tax. Accountant reviews, posts and sends it.
5. Record the customer's actual payment and reconcile the bank transaction. Confirm no overdue reminder remains eligible for the paid advance.
6. Complete the service or approve progress under the selected method. Create the remaining billing with the correct advance deduction. For a simple completed contract, net cumulative commercial invoicing should equal 30,000 MAD, not 36,000 MAD.
7. Record a partial payment of the final receivable. Verify the outstanding residual, due date and customer aging.
8. Create a justified 1,000 MAD tax-exclusive credit note for the test. Review its tax and analytic effect, and reconcile as appropriate. Net commercial invoicing becomes 29,000 MAD under this simplified example.
9. For an eligible overdue residual, run follow-up: create one appropriate reminder. Place the invoice on dispute hold and verify no further automated reminder is sent.
10. Import a bank transaction with ambiguous reference/amount and verify it stays for review. Reimport the statement and confirm no duplication.

Review results

Trace CRM → quotation → signed order → chantier → invoices/credit → payment/reconciliation. Confirm the dashboard's revenue, residual and conversion definitions match these records. Do not interpret invoiced service value minus the pilot's material cost as full project profit: labor, subcontracting and other costs are absent.

Closeout exercise

Try closing the site with outstanding procurement and unresolved financial exceptions from Pilot A. Closure should flag/block according to the approved policy. Resolve or explicitly approve exceptions; archive without losing document access and history. Record sales, stock and accounting owners' acceptance separately.

28 | Data migration and reconciliation

Decide the migration basis

Choose detailed historical transactions or controlled opening balances with accessible historical archives. Do not import history and opening balances for the same financial period/effect. Record the cutoff date and treatment of partially completed orders, outstanding bills, customer advances, stock and analytic history.

Import in dependency order

1. Companies, users/roles and required configuration.
2. Accounts, taxes, journals, payment terms and analytic plans under accountant control.
3. Contacts and addresses using stable external IDs.
4. Units, categories, products, vendor prices and tracking information.
5. Chantiers/projects, analytic accounts and locations with an explicit mapping.
6. Employees and restricted contracts/documents through an HR-controlled import.
7. Opening stock quantities by location/lot/serial, with an approved valuation method.
8. Open commercial/procurement documents and opening financial items under the chosen migration strategy.
9. Documents and links, then remaining historical reporting records if included.

Practical procedure

Export an Odoo template for each model and prepare a field mapping. Clean and validate a small sample first. Use the import test facility where available, correct rejected rows and run the real sample. Verify it in the UI and reports with the intended roles. Only then import the remaining data. Reimport updates with the same external IDs; do not rely on display names as unique keys.

Required reconciliations

- Contacts/products: source count, imported count, rejected count and deliberate merges.
- Stock: quantity by product/location/lot/serial and approved opening inventory value.
- Finance: trial balance, receivables, payables, bank/cash, advances and taxes against the approved opening position.
- Operations: open sales/PO quantities, fulfilled quantities and outstanding commitments.
- Analytics: project/account mapping, missing allocations and imported balances.
- HR/documents: record counts and confidentiality/access tests.

Opening stock can itself generate accounting valuation effects. Coordinate that with opening journal entries to avoid doubling inventory value. Use Odoo-supported import/ORM processes; do not insert transactional tables directly in SQL.

Acceptance gate

Business owners approve reconciliation totals and all unresolved exceptions. Freeze accepted mapping/configuration for the business cutoff. Retain source files, import IDs and correction logs. This page covers application data readiness; backup, server cutover and deployment procedures remain outside scope.

29 | Acceptance test catalogue

Required evidence matrix

ID	Scenario	Pass condition
SEC-01	Cross-company access via UI/export/link	Restricted records and aggregates stay inaccessible
SEC-02	HR attachment direct download	Unauthorized user denied
CORE-01	Initialize chantier twice	One project/account/location relationship
CRM-01	Send and repeat J+3 job	One eligible activity; none after closure
SAL-01	Portal signature and additional order	Correct customer scope; no duplicate site
SAL-02	Advance, final invoice and credit	Correct cumulative amount and residual
PUR-01	9,999 / 10,000 / 10,001 MAD	Exact signed threshold behavior
PUR-02	Self-approval and changed approved PO	Segregation and reapproval enforced
QUA-01	Partial pass/reject and bypass attempt	Only approved quantities released
STK-01	Partial receipt and supplier return	Correct quantities, backorder and source links
STK-02	Transfer, consumption, unused return	Physical and analytic totals reconcile
STK-03	Repeated/concurrent consumption	No duplicate effect or prohibited overconsumption
STK-04	Reorder with incoming stock/multiples	Correct procurement without approval bypass
FIN-01	Partial payment, fee, ambiguous match	Balanced entries and reviewed exceptions
FIN-02	Reimport statement	No duplicate transactions
FIN-03	Paid/disputed invoice follow-up	Suppressed; eligible reminder sent once
FIN-04	Reports and analytic dimensions	Ledger/balance/aging/allocations reconcile
HR-01	Leave approve/refuse/cancel	Correct balances and availability
HR-02	Recruit and appraise	Correct employee creation and confidentiality
DOC-01	Revision, search and expiry	Prior version retained; access and reminder correct
COM-01	External reply/internal note	Correct routing; no unintended external note
CAL-01	Reschedule/cancel/synchronize	Correct time, privacy and no duplicate
KPI-01	Eight metrics and drill-downs	Agreed definitions match source reports

For each test, record Odoo/addon versions, company, role, record references, steps, expected/actual result, evidence and defect owner. Failed financial, approval or confidentiality tests block their affected workflow. Retest the defect and relevant dependent behavior after correction; avoid repeating unrelated tests without a concrete reason.

30 | Phased implementation roadmap

Dependency-based delivery

Phase	Deliverables	Exit gate / owner
1. Foundation	Decisions, company, users, Contacts, accounting/product baseline	Access/master-data sample accepted by functional leads
2. Chantier core	Master, lifecycle, account/location links, migration mapping	CORE-01 and company checks; operations lead
3. Purchase/stock	Requests, approvals, receipts, quality, transfer/consumption/returns	Pilot A reconciles; buyer, stock, quality, accountant
4. CRM/sales	Pipeline, templates, pricing, portal, orders and billing	Customer cycle works; sales and accountant
5. Finance completion	Reports, matching, reminders, cheques and analytics	Pilot B and financial reconciliations; accountant
6. People/collaboration	HR, GED, Discuss and Calendar integrations	Privacy and workflow tests; HR/department owners
7. Dashboards	Eight KPIs, filters, drill-downs and performance checks	KPI sign-off; management
8. Business readiness	Imports, role training, procedures and exception closure	All mandatory requirements accepted

Configure basic email, Discuss and Calendar early so operational testing is realistic. Complete their integrated rules once the source processes exist. Start selecting Community gap addons in Phase 1; a late finance/GED compatibility discovery can change development scope.

First practical work package

1. Verify installed edition and addon inventory; record current custom models.
2. Create the configuration/decision register and confirm company/cost/approval assumptions.
3. Configure one company, six pilot roles, one supplier/customer and one material/service product pair.
4. Implement the chantier core and test duplicate/company controls.
5. Configure purchase/receipt routes and implement the quality gate.
6. Implement the transfer/consumption bridge and run Pilot A.
7. Correct demonstrated issues before scaling product imports or dashboard work.

Planning discipline

Estimate effort after the Community extension trials and existing-data assessment. Do not assign a precise completion date without knowing addon compatibility, data quality and business availability. Track progress by accepted workflows rather than installed module count. The critical dependency is trustworthy stock/accounting data, not the visual dashboard.

Acceptance gate

Every phase has a responsible business owner, required configuration/custom work, evidence and unresolved decisions. Later phases may prepare in parallel operationally, but must not declare results accepted while their financial/data prerequisites remain unresolved.

31 | Training, procedures and handover

Role-based training exercises

Audience	Hands-on exercise
Sales	Lead → site visit → quotation → signature → invoice request
Buyer/approver	Material request → supplier comparison → approval → bill exception
Storekeeper/quality	Partial receipt → inspection → release/reject → site transfer/return
Chantier manager	Request materials → receive → consume → review site balances
Accountant	Post → credit/partial payment → reconcile → report/remind
HR/employee	Submit/approve/cancel leave; restricted employee/appraisal workflow
Management	Read KPI definitions, filter, drill down and investigate variance

Standard operating procedures to deliver

Create short role-specific instructions using the configured menu labels and screenshots from the actual database. Include required inputs, who approves, how to correct errors, what is forbidden and where to report exceptions. Use the same test records as this guide so trainees can reconcile outcomes.

Ongoing application checks

Daily: review failed communications/jobs, pending approvals, quality holds, overdue receipts and unallocated draft financial items. Weekly: inspect unexplained stock differences, unresolved matching items, overdue collections and open site requests. At period end: reconcile inventory/accounting, review analytic completeness, settle exceptions and apply accountant-approved lock dates.

Handover package

- Configuration register with approved settings and ownership.
- Addon list with exact versions, licenses, dependencies and business purpose.
- Custom model/field and source-link documentation.
- Data-import mappings and reconciliation evidence.
- Test catalogue with results and known limitations.
- User procedures, approval matrix and exception contacts.
- KPI definitions and financial cost policy.
- Decision log showing any changed or deferred requirements.

Do not mark a requirement 'complete' when it is only installed, visually demonstrated by an administrator or replaced with a different capability without agreement. Keep the AI bank matching, centralized GED, appraisals and barcode-operation gaps visible until accepted solutions are demonstrated.

Business acceptance

Obtain process-owner confirmation that ordinary operations and corrections can be performed by the intended users. Record remaining low-impact issues with owner and target resolution. Financial integrity, confidentiality, approval bypasses and duplicate stock/accounting effects must be resolved before the affected process is used for real business.

32 | Decisions and requirement closure

Resolve during the foundation phase

Decision	Recommended starting point / required owner
Legal entities	One company for EL MOKRIF; add only real separate entities. Management.
Existing chantier model	Inventory existing records and choose one authoritative master. Developer/operations.
Purchase threshold	10,000 MAD from report; define tax basis, equality and currency policy. Management/accountant.
Approval separation	Buyer and approver distinct where mandated; changes trigger reapproval. Management.
Material costing	Internal transfers are not costs; define consumption attribution. Accountant/operations.
Sales billing	Materials vs services; advance/fixed/progress method by contract. Sales/accountant.
Quality	Inspection scope, pass criteria and rejected-goods process. Quality/buyer.
Taxes and HR	Validate current settings and employee entitlements; no hardcoded report assumptions. Accountant/HR.
AI matching	Define literal AI outcome, solution and review standard, or documented scope change. Finance/management.
Documents/barcodes/appraisals	Trial compatible extensions before custom build. Functional owner/developer.
History migration	Detailed history or reconciled opening balances, with cutoff. Accountant.
Multi-company dashboard	Authorized aggregation versus formal consolidation. Management/accountant.

Closure mapping to the source report

Configuration: roles/company tests. Discussion: routing/privacy tests. Calendar: event/integration tests. Accounting: invoice/payment/report/matching/cheque tests. HR: employee, contract, leave, recruitment and appraisal tests. Documents: centralized search/version/access tests. Contacts: data quality/import tests. Purchase: RFQ/agreement/approval/receipt tests. CRM: pipeline/target/activity tests. Inventory: traceability/quality/replenishment/valuation/scanning tests. Sales: template/pricing/signature/portal/billing tests. Dashboard: eight reconciled KPI tests.

Scope boundaries to retain

Payroll, statutory consolidation, sophisticated retention/progress certificates, manufacturing and equipment depreciation are not fully specified by this source. They require separate requirements when selected. Expenses, timesheets and detailed profitability are supporting options, not substitutes for required features. Deployment and infrastructure remain managed separately by the user.

Final readiness criterion

The system is ready for a workflow when its master data, permissions, standard configuration, custom logic, exception paths and reconciliation all pass with the actual roles. The plan is designed to make each of those results concrete and reviewable.

33 | Sources and implementation notes

Primary business requirements

Rapport_Odoo_ELMOKRIF (1) (1).pdf, 18 pages, supplied by the user. Principal sections used: 1.4 functional scope; 1.5 functional needs; 1.6 Community Edition requirement; 2.3 company/user organization; 2.4 sales and purchase flows; and Chapter 4 configuration of company, accounting, CRM/sales, stock/purchase, HR and dashboards. Deployment sections were excluded.

Odoo references

[1] Odoo editions comparison. Use to understand edition boundaries; the current comparison page is not a frozen 17.0 addon inventory. <https://www.odoo.com/page/editions>

[2] Odoo 17 down payments. Reference for the standard advance-invoice and deduction workflow. https://www.odoo.com/documentation/17.0/applications/sales/sales/invoicing/down_payment.html

[3] Odoo 17 pricelists. Reference for customer pricelists and advanced pricing rules. https://www.odoo.com/documentation/17.0/applications/sales/sales/products_prices/prices/pricing.html

[4] Odoo 17 automatic inventory valuation. Reference for category valuation settings and accounting integration. https://www.odoo.com/documentation/17.0/applications/inventory_and_mrp/inventory/product_management/inventory_valuation/inventory_valuation_config.html

[5] Odoo 17 Expenses. Reference for the optional expense-management foundation. <https://www.odoo.com/documentation/17.0/applications/finance/expenses.html>

[6] Odoo 17 online quotation signature (localized documentation). https://www.odoo.com/documentation/17.0/th/applications/sales/sales/send_quotations/get_signature_to_validate.html

[7] Official Odoo source repository. Inspect the 17.0 branch and installed addon manifests when implementing extensions. <https://github.com/odoo/odoo>

Evidence boundaries

The prior analysis consulted official documentation search results; some direct page/source retrievals were unavailable. This guide does not claim a full source-code audit or live installation verification. Proposed custom workflows, addon names and test designs are implementation recommendations derived from the report. Validate exact available fields, menus, methods and addon dependencies in the user's Odoo 17 installation before coding.

No current Moroccan tax or labor-law review was performed for this document. Rates, calendars, entitlements, cheque layouts and accounting policies must be approved by the responsible company professionals. No addon has been installed and no Odoo data has been changed as part of producing this guide.